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ELECTRICAL SUBMERSIBLE PUMPING SYSTEM WITH A RIDE THROUGH POWER SUPPLY HAVING A SAFETY INTERLOCK

Abstract

A well system includes an electrical submersible pumping (“ESP”) assembly, a variable speed drive, and a ride through system. The ESP assembly includes a pump for lifting liquid from inside the well and an electric motor for powering the pump. The variable speed drive is outside the well and controls delivery of electricity from a power source to the motor. The ride through system provides backup electricity if there is an interruption of electricity from the power source. The ride through system includes a bank of power modules having stored electrical power. The modules are connected in series and have a combined electrical potential adequate for powering the motor. By selectively disconnecting adjacent modules from one another, the electrical potential from the bank of power modules is reduced to a magnitude that is not hazardous for personnel.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of Invention

[0001] The present disclosure relates to an electrical submersible pumping system with a ride through power supply having a safety interlock.

2. Description of Prior Art

[0002] Electrical submersible pumping systems (“ESP systems”) are commonly used in hydrocarbon producing wells. Typically, ESP systems include an electrical motor, a pump having impellers and diffusers (which is usually an electrical submersible pump (“ESP”)), and a driveshaft connected between the motor and pump. The driveshaft is attached to the impellers, and the shaft and impellers rotate when the motor is energized. Rotating impellers draws fluid from inside the well into the pump and forces the fluid through the diffusers, which pressurizes the fluid for delivery to surface. When electrical power to the motor is lost at the same time a column of well fluid is above the impellers, fluid in the column is gravity driven downhole and across the impellers. The backflow often soon causes the impellers to rotate in a direction opposite their rotation during pressurization. The connected driveshaft also rotates to backspin the motor which generates electricity that is conducted up the well along a power cable connected from surface to the motor. This electricity creates undesired effects, such as posing a hazard for personnel operating the well. An additional undesired effect from backspin is motor damage, which can occur if the motor is reenergized during backspin.

[0003] As a corrective measure to address potential power losses, ride through or power through systems are sometimes installed in ESP systems, which temporarily supply electrical power to an ESP system during intermittent short term power outages. Typical ride through systems include a bank of batteries or super capacitors arranged in series to provide a voltage output necessary for short term electrical power supply for an ESP system. Examples of ride through (or power through) systems or methods are disclosed in U.S. Pat. Nos. 7,607,896, 8,092,190, and 11,451,170, which are assigned to the assignee of the present application and incorporated by reference herein in their entireties and for all purposes. Ride through systems are designed for driving an ESP motor, which results in the electrical potential stored in a ride through system be at a magnitude hazardous to operations personnel. Which introduces difficulties to maintenance personnel, such as the need to don personnel protective gear or drain the electricity from the ride through system, both of which are time consuming efforts that result in losses of revenue.

SUMMARY OF THE INVENTION

[0004] Disclosed herein is an example method of handling a ride through system for use with an electrical submersible pump (“ESP”) assembly, which includes disconnecting from one another a bank of power modules in the ride through system that are connected in series to reduce the electrical potential transmittable from the ride through system to lower than a lower threshold magnitude that is hazardous to personnel, and reconnecting the modules so that the electrical potential transmittable from the ride through system is at least at a higher threshold magnitude for powering the ESP assembly to oppose backspin. In an example, the bank of power modules are in a deactivated mode when sequential modules are disconnected from one another, and are in a standby mode when sequential modules are connected to one another. The modules are optionally serviced when the bank of power modules are in the deactivated mode. In an alternative, the step of disconnecting the sequential modules includes opening an electrical connection between the adjacent modules, which in one example, opening an electrical connection involves moving a switch into an open position, where the switch is in a line that selectively conducts electricity between the sequential modules. In an embodiment, each sequential power module is in selective

electrical communication by electrical connections that connect between the sequential power modules, and where the sequential modules are disconnected from one another at substantially the same time. In this example, an interlock is optionally coupled with each of the electrical connections and the method further includes activating the interlock to selectively connect and disconnect the sequential power modules. The ESP assembly is optionally selectively powered with electricity from a power source, and the modules are in selective electrical communication with the ESP assembly when electricity from the power source is suspended. Examples of the method include recharging the power modules.

[0005] Also disclosed is an example of a ride through system for use with an electrical submersible pump (“ESP”) assembly, which includes, power modules that each have a maximum electrical potential below a threshold magnitude that is hazardous to personnel and that selectively have a combined electrical potential which is at least at a threshold magnitude for powering the ESP by an amount that opposes backspin, and electrical connections between sequential power modules to define a bank of power modules that are arranged in series, the electrical connections selectively in a standby configuration so that a maximum electrical potential transmissible from the bank of power modules is substantially equal to a sum of the electrical potentials of each of the power modules, and the electrical connections selectively changeable into a deactivated mode in which the maximum electrical potential transmissible from the bank of power modules is substantially equal to an electrical potential of a one of the power modules. The ride through system optionally includes switches in the connections that are each in an open position when the electrical connections are in the deactivated mode and that are each in a closed position when the electrical connections are in the standby mode. In examples, an interlock is coupled to the switches, that selectively moves the switches between the open and closed positions. A controller is optionally included for monitoring electricity from an electrical source to the ESP assembly, and the interlock is in communication with the controller and is responsive to commands from the controller. Examples of power modules include a battery, a capacitor, a super capacitor, and combinations of these. Embodiments exist having one or more of, a variable speed drive for controlling electricity to the ESP assembly and that is in selective electrical communication with the bank of power modules, and a recharging power source for recharging the power modules. Optionally, the recharging power source is the same as a power source for providing electrical power to the ESP assembly. Example magnitudes of electrical potential in each of the power modules are at values that are substantially equal to one another, values that are different from one another, values that are substantially the same as others and different from others, and combinations.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] Some of the features and benefits of the present invention having been stated, others will become apparent as the description proceeds when taken in conjunction with the accompanying drawings, in which:

[0007] FIG. 1 is a side partial sectional view of an example of a well system that includes an electrical submersible pumping system and a variable speed drive.

[0008] FIG. 2A is a schematic example of a ride through system for use with the variable speed drive of FIG. 1, the ride through system being in a standby mode.

[0009] FIG. 2B is a schematic example of the ride through system of FIG. 2A in a deactivated mode.

[0010] While subject matter is described in connection with embodiments disclosed herein, it will be understood that the scope of the present disclosure is not limited to any particular embodiment. On the contrary, it is intended to cover all alternatives, modifications, and equivalents thereof.

DETAILED DESCRIPTION OF INVENTION

[0011] The method and system of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings in which embodiments are shown. The method and system of the present disclosure may be in many different forms and should not be construed as limited to the illustrated embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey its scope to those skilled in the art. Like numbers refer to like elements throughout. In an embodiment, usage of the term “about” includes $\pm 5\%$ of a cited magnitude. In an embodiment, the term “substantially” includes $\pm 5\%$ of a cited magnitude, comparison, or description. In an embodiment, usage of the term “generally” includes $\pm 10\%$ of a cited magnitude.

[0012] It is to be further understood that the scope of the present disclosure is not limited to the exact details of construction, operation, exact materials, or embodiments shown and described, as modifications and equivalents will be apparent to one skilled in the art. In the drawings and specification, there have been disclosed illustrative embodiments and, although specific terms are employed, they are used in a generic and descriptive sense only and not for the purpose of limitation.

[0013] In FIG. 1 is a side partial sectional view of an example of a well system **10** that includes an electrical submersible pumping (“ESP”) assembly **12** for artificially lifting liquid from within a well **14**, which intersects a formation **16**. ESP assembly **12** includes a pump **17** having impellers **18** and diffusers **19** stacked alternately between the impellers **18**. Rotating impellers **18** within the stationary diffusers **19** lifts liquid from within the well **14** upwards through production tubing **20** to surface. An inlet **21** is formed through a sidewall of the housing of pump **17**, which provides a path for the liquid to enter into pump **17** and into contact with the impellers **18** and diffusers **19**. The upper end of production tubing **20** terminates within a wellhead assembly **22** shown on surface. Inside wellhead assembly **22** liquid in production tubing **20** is transferred into a production line **24** shown extending laterally from wellhead assembly **22**. In alternatives, production line **24** transports fluid distally from wellsite. ESP assembly **12** also includes an electrical motor **26**, a connected shaft **27** shown in dashed outline having an end rotationally affixed to motor **26**, and an opposing end coupled with the impellers **18**. A seal section **28** for equalizing pressure within motor **26** is included with assembly **12**, which is shown between motor **26** and pump **17** and intersected by shaft **27**.

[0014] Well system **10** includes an electrical supply system **30** for providing electricity to the motor **26**. For the purposes of discussion herein, in an example electricity means a flow of electrons and measurable in terms of electrical current, and having a potential energy measurable in terms of voltage. Supply system **30** includes a power source **32**, which in examples includes alternating current and optionally provides three-phase power for driving motor **26**. Examples of power source **32** include an electrical utility, a generator, and a renewable source. In the example of FIG. 1, line **34** connects power source **32** to a motor controller **36**, such as a variable speed drive or other similar device. Examples of variable speed drive **36** include a pulse with modulated variable frequency drive, or other known controllers for varying current or voltage within the electricity being delivered from power source **32** which is used for varying the rotational speed of motor **26**. In the example shown, power source **32** and variable speed drive **36** are on surface and outside of wellbore **14**. A controller **38** is shown included with variable speed drive **36** and which in examples monitors characteristics of the electricity provided from power source **32** and includes means for adjusting operation of variable speed drive **36** in response to changes in characteristics of the electricity provided through power source **32**. Shown in dashed outline within variable speed drive **36** is a ride-through system **40**, which as explained in more detail below, includes stored electrical energy for providing power to motor **26** in instances when electricity from power source **32** is temporarily suspended or otherwise unavailable. A line **42** is shown electrically connecting variable speed drive **36** to a transformer **44**. A power cable **46** extends from an output of transformer **44**, through wellhead assembly **22** and into wellbore **14**. An end of power cable **46** opposite

transformer **44** connects to a splice **48**, which provides connection to a motor lead extension **50** (which is optionally part of the power cable **46**). Motor lead extension **50** terminates at a pothead connector **52** shown mounted onto motor **26** and provides electrical connection from the motor lead extension **50** to the motor **26**. Further shown in FIG. **1** is that wellbore **14** is lined with casing **54**, perforations **56** extend radially through the sidewalls of casing **54** that permit the flow of fluid **F** from within formation **16** and to within inside of casing **54**.

[0015] Shown in FIG. **2A** is a schematic example of ride-through system **40**. Electrically conducting leads **58**, **60** are on opposing ends of ride-through system **40**, and provide electrical communication between ride-through system **40** and variable speed drive **36**. The ride-through system **40** of FIG. **2A** includes power modules **62.sub.1-n**. Examples of the power modules **62.sub.1-n** include batteries, capacitors, super capacitors, or any device capable of retaining a designated amount of stored electrical energy for a particular period of time. The modules **62.sub.1-n** each have a positive lead (+) and a negative lead (-), and during the particular period of time when the electrical energy is stored in the modules **62.sub.1-n**, an electrical potential exists between the positive and negative leads. In a non-limiting example, the electrical energy stored in the modules **62.sub.1-n**, which is alternatively referred to as power capacity and is expressed in terms of Watt-hours. In the illustrated embodiment the modules **62.sub.1-n** are arranged in series so that the electrical potential difference between lead **58** and lead **60** is substantially equal to the combined electrical potentials across each one of the modules **62.sub.1-n** (i.e., across their respective positive and negative leads), and which defines the electrical potential of the ride-through system **40**. Further illustrated in FIG. **2A** is that a bank of modules **63** is defined by the power modules **62.sub.1-n** connected in series. In alternatives (not shown), the power modules **62.sub.1-n** are arranged in a combination of parallel and series configurations.

[0016] Connector lines **64.sub.1-n** extend between sequential ones of the power modules **62.sub.1-n** and within each of lines **64.sub.1-n** are switches **66.sub.1-n** that are shown in a closed configuration in FIG. **2A**. For the purposes of discussion herein, sequential or arranged sequentially defines an arrangement in which the power modules **62.sub.1-n** are connected in an electrical circuit, e.g., end to end, such that terminals on the modules **62.sub.1-n** connect to terminals having a different polarity. Charging power sources **68.sub.1-n** are shown connected across each of the power modules **62.sub.1-n** via connector lines **70.sub.1-n**, **72.sub.1-n** that provide electrical communication to the positive and negative terminals respectively of each of the power modules **62.sub.1-n**. Optionally, recharging power sources **68.sub.1-n** are made up of a single power source (not shown) that connects to the power modules **62.sub.1-n** along connector lines **70.sub.1-n**, **72.sub.1-n**; optionally, the single power source is the power source **32**. In the example of FIG. **2A**, the switches **66.sub.1-n** being in the closed configuration is referred to as a standby mode. When in the standby mode and a loss of power or drop of power from power source **32** is sensed by controller **38**, internal switching (not shown) provides electrical communication between ride-through module **40** and line **42**, so that electricity flows from the modules **62.sub.1-n** in the ride-through module **40** to the ESP assembly **12** (FIG. **1**) via line **42**. Further in this example, the electrical energy stored in modules **62.sub.1-n** provides a threshold magnitude of electrical potential and current to motor **26** and causing motor **26** to exert a rotational torque to the shaft **27** and attached impellers **18**. In an embodiment, the rotational torque is applied to the impellers **18** in the same rotational direction for lifting liquid by operation of the pump **17** ("pumping direction"), which is opposite to the rotational direction of the impellers **18** occurring during backspin ("backspin direction"), and the magnitude of the rotational torque is at least that of a rotational torque applied to the impellers **18** from a static head of liquid in and above the pump **17**. Exerting a rotational torque against the impellers **18** in this direction and magnitude opposes and/or prevents backspin as the impellers **18** are either maintained in a rotationally stationary position or are rotated in the pumping direction. Further embodiments exist in which the rotational torque from the motor **26** while being energized by the ride through module **40**, rotates the impellers **18** at a rate that

prevents backflow.

[0017] Still referring to FIG. 2A, an interlock 74 is shown having legs coupled with each of the switches 66.sub.1-n, as described in more detail below interlock 74 provides a means for reconfiguring switches 66.sub.1-n from the closed position shown in FIG. 2A and into an open configuration. Communication means 76, which in examples is wireless, hard-wired, or fiber optics, provides signal communication between controller 38 and interlock 74. Referring now to FIG. 2B, a schematic example of the ride-through system 40 is shown in a deactivated mode. In the deactivated mode the switches 66.sub.1-n are in an open configuration so that each of the power modules 62.sub.1-n are disconnected and electrically isolated from one another. In a non-limiting example, the switches 66.sub.1-n are moved into the open position by operation of the interlock 74. In alternatives, the interlock 74 operates in response to receiving an activation signal from controller 38 via communication means 76. In the deactivated mode, and with sequential power modules 62.sub.1-n electrically isolated from one another, the maximum electrical potential transmittable from the ride-through system 40 is no greater than the maximum electrical potential available from any one of the power modules 62.sub.1-n, that as across their respective positive and negative terminals. In examples, electricity that is transmittable is that which is conducted, or able to be conducted. In embodiments, the maximum electrical potential in any one of the power modules 62.sub.1-n is at or below a threshold magnitude that is hazardous to personnel, which in one example is about 50 V. In examples when the ride-through module 40 is in the deactivated configuration, operations personnel freely handle the ride-through module 40 without personal protection equipment and without draining electrical energy from the power modules 62.sub.1-n.

[0018] In a non-limiting example of operation, electricity from power source 32 (FIG. 1) is received by variable speed drive 36, where it in combination with commands from controller 38, condition the electricity for use by motor 26 to power the pump 17 so that liquid inside well 14 is lifted to wellhead assembly 22. When the supply of electricity from power source 32 is interrupted, which in examples occurs for a period measured in milli-seconds to in excess of many minutes, ride-through system 40 of FIG. 2A is put into electrical communication with ESP assembly 12 for temporarily providing electricity to motor 26. Upon resumption of the supply of electricity from power source 32, ride-through system 40 is disconnected from ESP assembly 12 and one or more of power modules 62.sub.1-n are recharged using recharging power source(s) 68.sub.1-n. While power modules 62.sub.1-n have a substantially full charge, the interlock 74 is operated to reconfigure the ride-through system 40 from the standby mode of FIG. 2A to the deactivated mode shown in FIG. 2B. When in the deactivated mode, maintenance, service, repair, or replacement of all or a part of the ride-through system 40 is performed. Further in this example, upon completion of service, maintenance, or other handling by operations personnel of the ride-through module 40 when in the deactivated mode, a signal is generated from controller via communication means to interlock 74 to reconfigure. switches 66.sub.1-n into the standby mode of FIG. 2A, which in a loss of power, as monitored through controller 38, motor 26 is optionally powered from electricity from the ride-through system 40. In alternatives, the interlock 74 moves switches 66.sub.1-n between the open and closed position simultaneously, or at different times, such as in a particular sequence.

[0019] . In embodiments, controller 38 includes a processor and a non-transitory computer-readable memory accessible by the processor and having executable code stored thereon. The executable code includes a set of instructions that causes a processor to perform operations that include operating the interlock 74, adjusting operation of the variable speed drive 36, and connecting ride-through system 40 with the ESP assembly 12. Examples of processor include a multicore processor with nodes such as those from Intel Corporation or Advanced Micro Devices (AMD), or an HPC Linux cluster computer. Process optionally is a mainframe computer of any conventional type of suitable processing capacity such as those available from International Business Machines (IBM) of Armonk, N.Y. or other source; or alternatively is a computer of any conventional type of suitable processing capacity, such as a personal computer, laptop computer, or any other suitable processing

apparatus. It should thus be understood that a number of commercially available data processing systems and types of computers may be used for this purpose.

[0020] The present invention described herein, therefore, is well adapted to carry out the objects and attain the ends and advantages mentioned, as well as others inherent therein. While a presently preferred embodiment of the invention has been given for purposes of disclosure, numerous changes exist in the details of procedures for accomplishing the desired results. These and other similar modifications will readily suggest themselves to those skilled in the art, and are intended to be encompassed within the spirit of the present invention disclosed herein and the scope of the appended claims.

Claims

1. A method of handling a ride through system for use with an electrical submersible pump (“ESP”) assembly, the method comprising: disconnecting from one another a bank of power modules in the ride through system that are connected in series to reduce the electrical potential transmittable from the ride through system to lower than a lower threshold magnitude that is hazardous to personnel; and reconnecting the modules so that the electrical potential transmittable from the ride through system is at least at a higher threshold magnitude for powering the ESP assembly to oppose backspin.
2. The method of claim 1, wherein the bank of power modules are in a deactivated mode when sequential modules are disconnected from one another, and are in a standby mode when sequential modules are connected to one another.
3. The method of claim 2, further comprising servicing the modules when the bank of power modules are in the deactivated mode.
4. The method of claim 1, wherein the step of disconnecting the sequential modules comprises opening an electrical connection between the adjacent modules.
5. The method of claim 4, wherein opening an electrical connection comprises moving a switch into an open position, wherein the switch is in a line that selectively conducts electricity between the sequential modules.
6. The method of claim 1, wherein each sequential power module is in selective electrical communication by electrical connections that connect between the sequential power modules, and wherein the sequential modules are disconnected from one another at substantially the same time.
7. The method of claim 6, wherein an interlock is coupled with each of the electrical connections, the method further comprising activating the interlock to selectively connect and disconnect the sequential power modules.
8. The method of claim 1, wherein the ESP assembly is selectively powered with electricity from a power source, and wherein the modules are in selective electrical communication with the ESP assembly when electricity from the power source is suspended.
9. The method of claim 8, further comprising recharging the power modules.
10. A ride through system for use with an electrical submersible pump (“ESP”) assembly, the ride through system comprising: power modules that each have a maximum electrical potential below a threshold magnitude that is hazardous to personnel and that selectively have a combined electrical potential which is at least at a threshold magnitude for powering the ESP by an amount that opposes backspin; and electrical connections between sequential power modules to define a bank of power modules that are arranged in series, the electrical connections selectively in a standby configuration so that a maximum electrical potential transmissible from the bank of power modules is substantially equal to a sum of the electrical potentials of each of the power modules, and the electrical connections selectively changeable into a deactivated mode in which the maximum electrical potential transmissible from the bank of power modules is substantially equal to an electrical potential of a one of the power modules.

- 11.** The ride through system of claim 10, further comprising switches in the connections that are each in an open position when the electrical connections are in the deactivated mode and that are each in a closed position when the electrical connections are in the standby mode.
 - 12.** The ride through system of claim 11, further comprising an interlock coupled to the switches, that selectively moves the switches between the open and closed positions.
 - 13.** The ride through system of claim 12, further comprising a controller for monitoring electricity from an electrical source to the ESP assembly, and wherein the interlock is in communication with the controller and is responsive to commands from the controller.
 - 14.** The ride through system of claim 10, wherein the power modules are selected from the group consisting of a battery, a capacitor, a super capacitor, and combinations.
 - 15.** The ride through system of claim 10, wherein a variable speed drive for controlling electricity to the ESP assembly is in selective electrical communication with the bank of power modules.
 - 16.** The ride through system of claim 10, further comprising a recharging power source for recharging the power modules.
 - 17.** The ride through system of claim 16, wherein the recharging power source is the same as a power source for providing electrical power to the ESP assembly.
 - 18.** The ride through system of claim 10, wherein the magnitude of electrical potential in each of the power modules are at values selected from the group consisting of values that are substantially equal to one another, values that are different from one another, values that are substantially the same as others and different from others, and combinations.
 - 19.** The ride through system of claim 14, wherein the power modules are comprised of an arrangement of series and parallel configurations.
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